Total No. of Questions : 6]

SEAT No. :

P5044

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TE/Insem - 542
T.E. (Information Technology) (Semester - I)
THEORY OF COMPUTATION
(2012 Pattern)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4. Q5 or Q.6.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Assume suitable data if necessary.*

Q1) a) Define the following with suitable example. **[5]**

i) Moore machine

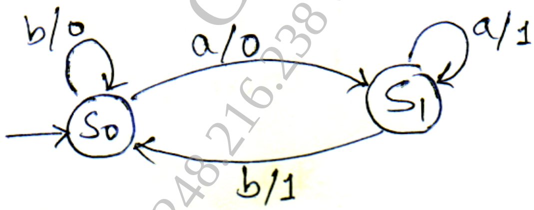
ii) Nondeterminstic finite automata

b) Design a DFA for the language over $\Sigma=\{0,1\}$ containing words ending with '101' **[5]**

OR

Q2) a) State in brief finite state machine properties and its limitations. **[5]**

b) Convert the following mealy machine to its equivalent moore machine. **[5]**

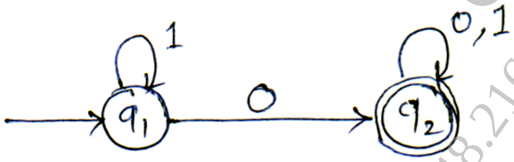


Q3) a) Define the following with suitable example **[5]**

i) Regular grammar

ii) Regular expression.

b) Obtain regular expression for the give FA using Arden's theorem. **[5]**



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OR

Q4) a) State pumping lemma for regular language. Also state closure properties of regular language. **[5]**

b) Construct equivalent FA for the following regular expression. **[5]**

$$1^* \, 00(01)^*$$

Q5) a) Consider following CFG : G = {(S,A), (a,b), P,S} where P consists of

$S \rightarrow aAS/a$

$A \rightarrow SbA/SS/ba$

Derive string "aabbaa" using leftmost derivation and rightmost derivation. Draw the derivation tree for the string. **[5]**

b) Define with suitable example **[5]**

i) Ambiguous grammar

ii) Greibach normal form

OR

Q6) a) Express following grammar using CNF. **[6]**

$S \rightarrow ABA$

$A \rightarrow aA|\in$

$B \rightarrow bB|\in$

b) Compare context-free grammar and regular grammar. **[4]**

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